Math 541

Homework 4

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Exercise 1. *The Klein bottle K is the space obtained from a square by means of the labelling scheme $aba^{-1}b$. Describe a two sheeting covering map $p : T^2 \to K$, where T^2 is the torus.*

Proof. Take the torus as a rectangle with opposite sides identified, and cut it into two horizontal strips. Map each strip homeomorphically onto the square for the Klein bottle; map one strip directly, and map the other after a horizontal flip. On the boundary, the side identifications match exactly the Klein bottle description $aba^{-1}b$. Hence this gives a 2-sheeted covering $T^2 \to K$. □

Exercise 2. *What compact surfaces cover the Klein bottle? Justify your answer.*

Proof. Supppose $p : E \to K$ is an n-sheeted coveirng space of the Klein bottle. Then $\chi(E) = n \cdot \chi(K) = n \cdot 0 = 0$. It must be the case that E is either a Torus or a Klein bottle. □

Exercise 3. *Let N_g be a compact non-orientable surface of genus $g > 1$. We are given that there exists a two sheeted covering map $p : S \to N_g$, where S is a compact orientable surface. What is the genus of S? Justify your answer.*

Proof. Let the genus of S be h. Since $p : S \to N_g$ is a 2-sheeted covering map, we have that $\chi(S) = 2\chi(N_g)$. Since N_g is the compact non-orientable surface of genus g, we have $\chi(N_g) = 2 - g$. Hence $\chi(S) = 2(2 - g) = 4 - 2g$. Since S is compact and orientable of genus h, we have $\chi(S) = 2 - 2h$. Equating the two formulas for $\chi(S)$ gives $2 - 2h = 4 - 2g$. Solving for h gives that the genus of S is $g - 1$. □

Exercise 4. *Let S_3 denote the compact orientable surface of genus 3. Assume that, for every positive integer n there exists an n-sheeted connected covering space of S^3. State which compact surfaces cover S_3. Justify your answer.*

Proof. Let $p : E \to S_3$ be an n-sheeted connected covering, where S_3 is the compact orientable surface of genus 3. Since covering spaces of orientable surfaces are orientable, E is also a compact orientable surface. Now $\chi(S_3) = 2 - 2(3) = -4$. For an n-sheeted covering, $\chi(E) = n\chi(S_3) = -4n$. Since E is compact and orientable, E has some genus h, and $\chi(E) = 2 - 2h$. Equating the two gives $2 - 2h = -4n$. Solving for h tells us that every connected compact surface covering S_3 is the orientable surface of genus $2n + 1$. □

Exercise 5. *Prove that a non-trivial finite group is not a free group.*

Proof. Let F be a free group. We will show that every nonidentity element in F has infinite order, establishing that F must be an infinite group. Suppose $x_1^{\delta_1} \dots x_m^{\delta_n}$ is a nonidentity element with order $k > 1$. Then:

$$\underbrace{(x_1^{\delta_1} \dots x_n^{\delta_n}) \dots (x_1^{\delta_1} \dots x_n^{\delta_n})}_{\text{k-times}} = 1.$$

For this to be true, it must be the case that $x_n^{\delta_n} x_1^{\delta_1} = 1$. In particular, we have:

$$x_1^{\delta_1}(x_2^{\delta_2} \dots x_{n-1}^{\delta_{n-1}})(x_1^{\delta_1})^{-1} = x_1^{\delta_1} \dots x_n^{\delta_n}.$$

Since $x_2^{\delta_2} \dots x_{n-1}^{\delta_{n-1}}$ is conjugate with $x_1^{\delta_1} \dots x_n^{\delta_n}$, they have the same order. We continue this process inductively, and ultimately show that $x_i^{\delta_i} x_{n-i+1}^{\delta_{n-i+1}} = 1$. Since inverses commute, we can reduce our original word down to the identity 1 from the inside out. This contradict our claim that $x_1^{\delta_1} \dots x_n^{\delta_n}$ is a nonidentity element.

□

Exercise 6. *Use the universal property of free products to prove that $\mathbb{Z} * (\mathbb{Z}/n\mathbb{Z})$ is non-abelian, for every integer n greater than 1.*

Proof. Let $\varphi_{\mathbb{Z}} : \mathbb{Z} \to \mathbb{Z} * \mathbb{Z}/n\mathbb{Z}$ and $\varphi_{\mathbb{Z}/n\mathbb{Z}} : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z} * \mathbb{Z}/n\mathbb{Z}$ be the canonical maps. Let $\sigma := (0, 1) \in S_{n+1}$ and $\tau := (123 \dots n) \in S_{n+1}$. Note that we can find group homomorphisms $\Phi_{\mathbb{Z}} : \mathbb{Z} \to S_{n+1}$ and $\Phi_{\mathbb{Z}/n\mathbb{Z}} : \mathbb{Z}/n\mathbb{Z} \to S_{n+1}$ such that $\Phi_{\mathbb{Z}}(1) = \sigma$ and $\Phi_{\mathbb{Z}/n\mathbb{Z}}(1) = \tau$. By the universal property of free products, there exists a map $f : \mathbb{Z} * \mathbb{Z}/n\mathbb{Z} \to S_{n+1}$ such that $f \circ \varphi_{\mathbb{Z}} = \Phi_{\mathbb{Z}}$ and $f \circ \varphi_{\mathbb{Z}/n\mathbb{Z}} = \Phi_{\mathbb{Z}/n\mathbb{Z}}$. In particular, $f(\varphi_{\mathbb{Z}}(1)) = \sigma$ and $f(\varphi_{\mathbb{Z}/n\mathbb{Z}}(1)) = \tau$. Since σ and τ do not commute, it must be the case that $\varphi_{\mathbb{Z}}(1)$ and $\varphi_{\mathbb{Z}/n\mathbb{Z}}(1)$ do not commute; i.e., $\mathbb{Z} * \mathbb{Z}/n\mathbb{Z}$ is not abelian. □

Exercise 7.

(a) Let H and K be conjugate subgroups of a group G. Prove that if f is any homomorphism of G into some other group such that $f(H) = 1$, then $f(K) = 1$ also.

(b) Let $G = A * B$, where A and B are nontrivial groups. Prove that A and B are not conjugate. [Hint: Use part (a) and the universal property of free products.]

Proof. (a) Since H and K are conjugate subgroups, there exists some $g \in G$ such that $K = gHg^{-1}$. Now let $k \in K$. Then $k = ghg^{-1}$ for some $h \in H$. Applying f gives $f(k) = f(g)f(h)f(g)^{-1}$. But because $f(H) = 1$, we have $f(h) = 1$. So $f(k) = f(g)\,1\,f(g)^{-1} = 1$. This holds for every $k \in K$, so $f(K) = 1$.

(b) Suppose towards contradiction that A and B are conjugate subgroups of $A * B$. Define homomorphisms $\alpha : A \to B$ and $\beta : B \to B$ such that $\alpha(a) = 1$ and $\beta(b) = b$. By the universal property of free products, there is a homomorphism $f : A * B \to B$ such that $f|_A = \alpha$ and $f|_B = \beta$. In particular, $f(A) = 1$. Since A and B are assumed conjugate, by part (a) this implies that $f(B) = 1$. But $f|_B = \mathrm{id}_B$, so $f(B) = B$. Hence $B = 1$. This is a contradiction, as we assumed B is nontrivial. Thus A and B are not conjugate.

□